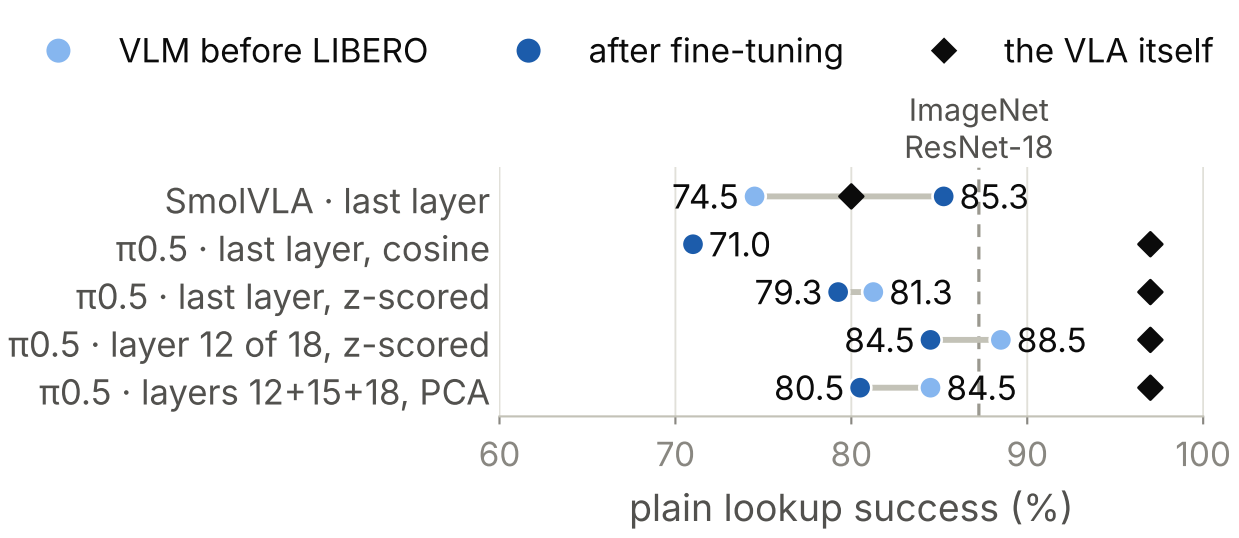
VLM before LIBERO
after fine-tuning
the VLA itself
ImageNet
ResNet-18
SmolVLA · last layer
π0.5 · last layer, cosine
π0.5 · last layer, z-scored
π0.5 · layer 12 of 18, z-scored
π0.5 · layers 12+15+18, PCA
74.5
85.3
71.0
79.3
81.3
84.5
88.5
80.5
84.5
60
70
80
90
100
plain lookup success (%)